



Hospital Management Sytem

Enhancing Health Systems for a Brighter Future

DEVELOPED BY:
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PROJECT DESCRIPTION:

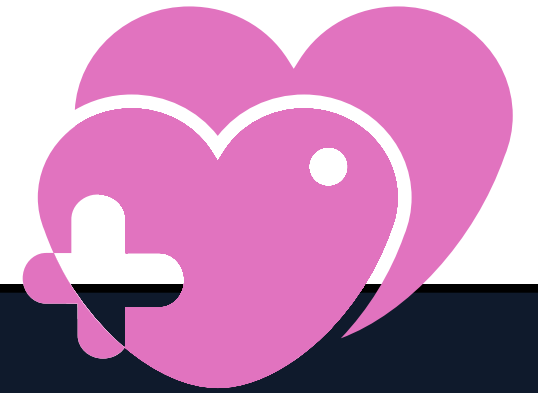
The Hospital Management System is a comprehensive software application developed to automate and manage all hospital operations efficiently. It handles patient registration, appointments, billing, medical records, staff management, and inventory control. The system ensures smooth coordination between departments, reduces paperwork, and improves service quality. It aims to enhance patient care while optimizing administrative and clinical workflows



PROJECT

AIM:

- To automate and streamline all hospital operations and processes
- To improve the quality and efficiency of patient care
- To ensure accurate and secure management of patient and hospital data
- To enhance communication and coordination among hospital departments
- To reduce manual work, errors, and administrative overhead
- To support data-driven decision-making and reporting



TABLES:



PATIENTS:

PatientID	FirstName	LastName	Gender	DateOfBirth	ContactNumber	Address
1	John	Doe	Male	1985-06-15	1234567890	123 Main St
2	Jane	Smith	Female	1990-02-10	9876543210	456 Oak Ave
3	Alice	Johnson	Female	1978-09-25	6541237890	789 Pine Rd
4	Bob	Williams	Male	1965-11-12	3216549870	321 Maple Dr
5	Emily	Brown	Female	1988-03-08	7412589630	111 Elm St
6	Michael	Davis	Male	2000-07-19	3698521470	333 Cedar Blvd
7	Sarah	Miller	Female	1995-05-30	8523697410	222 Birch Ln
8	David	Wilson	Male	1982-12-03	9631478520	444 Spruce Ct
9	Laura	Taylor	Female	1975-01-14	7419638520	555 Willow Way
10	Kevin	Anderson	Male	1992-10-20	7531598420	666 Aspen St

DOCTORS:

DoctorID	FirstName	LastName	Specialization	ContactNumber	Email	Schedule
1	Dr. Emma	Clark	Cardiology	1112223333	emma.clark@hospital.com	Mon-Wed-Fri 10a...
2	Dr. Liam	Scott	Neurology	2223334444	liam.scott@hospital.com	Tue-Thu 1pm 5pm
3	Dr. Olivia	Green	Orthopedics	3334445555	olivia.green@hospital.com	Mon-Fri 9am-12pm
4	Dr. Noah	Adams	Pediatrics	4445556666	noah.adams@hospital.com	Wed-Sat 10am-4pm
5	Dr. Ava	Baker	Dermatology	5556667777	ava.baker@hospital.com	Mon-Fri 2pm 6pm
6	Dr. William	Hall	ENT	6667778888	william.hall@hospital.com	Tue-Thu 10am 3pm
7	Dr. Sophia	Allen	Gynecology	7778889999	sophia.allen@hospital.com	Mon-Wed Fri 1pm...
8	Dr. James	Wright	General Medicine	8889990000	james.wright@hospital.com	Daily 8am-1pm
9	Dr. Mia	King	Psychiatry	9990001111	mia.king@hospital.com	Tue-Fri 11am-4pm
10	Dr. Benjamin	Lee	Radiology	0001112222	benjamin.lee@hospital.com	Mon-Thu 9am-1pm

ADMISSIONS:

AdmissionID	PatientID	RoomNumber	AdmissionDate	DischargeDate	Diagnosis
1	1	101A	2025-06-28	2025-07-01	Chest Pain
2	2	102B	2025-06-29	2025-07-02	Migraine
3	3	103C	2025-06-30	NULL	Fracture
4	4	104A	2025-07-01	2025-07-03	Flu
5	5	105B	2025-07-02	NULL	Skin Allergy
6	6	106C	2025-07-03	NULL	Ear Infection
7	7	107A	2025-07-04	NULL	Pregnancy Monitoring
8	8	108B	2025-07-05	NULL	Routine Checkup
9	9	109C	2025-07-06	NULL	Depression
10	10	110A	2025-07-07	NULL	X-ray Required

APPOINTMENTS:

AppointmentID	PatientID	DoctorID	AppointmentDate	AppointmentTime	Status
1	1	1	2025-07-01	10:00:00	Completed
2	2	2	2025-07-02	13:30:00	Scheduled
3	3	3	2025-07-03	09:15:00	Cancelled
4	4	4	2025-07-04	10:45:00	Completed
5	5	5	2025-07-05	15:00:00	Scheduled
6	6	6	2025-07-06	11:00:00	Completed
7	7	7	2025-07-07	14:20:00	Scheduled
8	8	8	2025-07-08	08:30:00	Completed
9	9	9	2025-07-09	12:00:00	Scheduled
10	10	10	2025-07-10	09:45:00	Scheduled

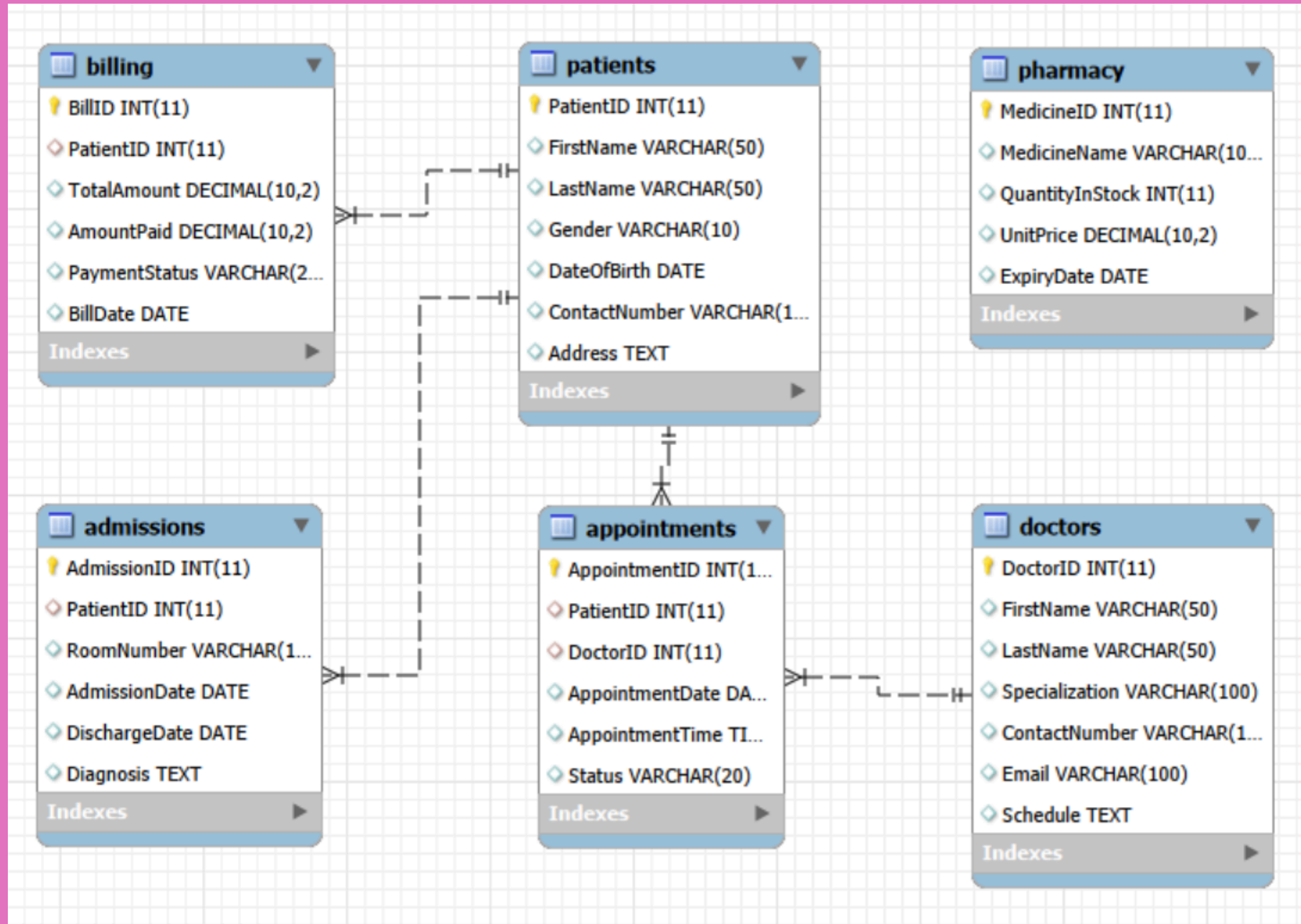
BILLING:

BillID	PatientID	TotalAmount	AmountPaid	PaymentStatus	BillDate
1	1	1200.00	1200.00	Paid	2025-07-01
2	2	950.00	950.00	Paid	2025-07-02
3	3	1500.00	1000.00	Partial	2025-07-03
4	4	800.00	800.00	Paid	2025-07-04
5	5	600.00	0.00	Unpaid	2025-07-05
6	6	1100.00	1100.00	Paid	2025-07-06
7	7	700.00	700.00	Paid	2025-07-07
8	8	500.00	500.00	Paid	2025-07-08
9	9	1300.00	0.00	Unpaid	2025-07-09
10	10	900.00	900.00	Paid	2025-07-10

PHARMACY:

MedicineID	MedicineName	QuantityInStock	UnitPrice	ExpiryDate
1	Paracetamol 500mg	200	1.50	2026-12-31
2	Amoxicillin 250mg	150	2.00	2026-10-30
3	Ibuprofen 400mg	100	1.75	2026-08-15
4	Cough Syrup	80	3.50	2026-09-20
5	Antacid Tablets	250	1.20	2026-11-10
6	Vitamin C 1000mg	120	2.80	2027-01-01
7	Insulin Injection	50	25.00	2026-07-31
8	Eye Drops	60	4.00	2026-06-30
9	Antibiotic Cream	40	5.25	2026-05-25
10	Pain Relief Spray	30	6.00	2026-08-01

ERR DIAGRAM:



DATA ANALYSIS:



BASIC FUNCTIONS

- **List of all male patients born after 1980.**

Syntax:

```
SELECT firstname, lastname, gender, dateofbirth FROM  
patients  
WHERE year(dateofbirth) > 1980  
AND gender = 'Male';
```

firstname	lastname	gender	dateofbirth
John	Doe	Male	1985-06-15
Michael	Davis	Male	2000-07-19
David	Wilson	Male	1982-12-03
Kevin	Anderson	Male	1992-10-20

The total number of male patients born after 1980 is 4.

- List of rooms that are empty along with the patient diagnosis and discharge date.

Syntax:

```
SELECT roomnumber, dischargedate, diagnosis FROM admissions  
WHERE dischargedate IS NOT NULL;
```

roomnumber	dischargedate	diagnosis
101A	2025-07-01	Chest Pain
102B	2025-07-02	Migraine
104A	2025-07-03	Flu

There are 3 rooms empty in the hospital after the patients are discharged.

AGGREGATE FUNCTIONS

- Calculate total revenue received (sum of amount paid).

Syntax:

```
SELECT SUM(AmountPaid) AS TotalRevenue  
FROM Billing;
```

TotalRevenue
7150.00

The Total Revenue of the Hospital is Rs. 7150.00.

- Find the most purchased medicine by the hospital along with it's total cost.

Syntax:

```
SELECT MedicineName, max(quantityinstock) AS TotalStock, quantityinstock*unitprice AS  
TotalCost from Pharmacy;
```

MedicineName	TotalStock	TotalCost
Paracetamol 500mg	250	300.00

Paracetamol 500mg was the most bought medicine by the hospital with total price of Rs.300.00.

JOIN FUNCTIONS

- List all appointments with patient and doctor names.

Syntax:

```
SELECT a.AppointmentDate, a.AppointmentTime, p.FirstName AS PatientName, d.FirstName AS  
DoctorName FROM Appointments a  
JOIN Patients p ON a.PatientID = p.PatientID  
JOIN Doctors d ON a.DoctorID = d.DoctorID;
```

AppointmentDate	AppointmentTime	PatientName	DoctorName
2025-07-01	10:00:00	John	Dr. Emma
2025-07-02	13:30:00	Jane	Dr. Liam
2025-07-03	09:15:00	Alice	Dr. Olivia
2025-07-04	10:45:00	Bob	Dr. Noah

- **Show appointments with doctor specialization.**

Syntax:

```
SELECT p.FirstName AS PatientName,  
d.FirstName AS DoctorName,  
d.Specialization,  
a.AppointmentDate  
FROM Appointments a  
JOIN Patients p ON a.PatientID = p.PatientID  
JOIN Doctors d ON a.DoctorID = d.DoctorID;
```

PatientName	DoctorName	Specialization	AppointmentDate
John	Dr. Emma	Cardiology	2025-07-01
Jane	Dr. Liam	Neurology	2025-07-02
Alice	Dr. Olivia	Orthopedics	2025-07-03
Bob	Dr. Noah	Pediatrics	2025-07-04

SUBQUERIES

- Find the name of the doctor with the highest number of appointments.

Syntax:

```
SELECT FirstName, LastName  
FROM Doctors  
WHERE DoctorID = (  
  SELECT DoctorID  
  FROM Appointments  
  GROUP BY DoctorID  
  ORDER BY COUNT(*) DESC  
  LIMIT 1);
```

FirstName	LastName
Dr. Sophia	Allen

Dr. Sophia Allen has the most number of appointments scheduled.

- **Find all doctors who have appointments with female patients.**

Syntax:

```
SELECT DISTINCT FirstName, LastName FROM Doctors
WHERE DoctorID IN (
SELECT a.DoctorID
FROM Appointments a
JOIN Patients p ON a.PatientID = p.PatientID
WHERE p.Gender = 'Female');
```

FirstName	LastName
Dr. Liam	Scott
Dr. Olivia	Green
Dr. Ava	Baker
Dr. Sophia	Allen
Dr. Mia	King

There are 5 doctors who have scheduled appointments with female patients

WINDOWS FUNCTIONS

- Sort the Amount payable by the patients in descending order and rank them.

Syntax:

```
SELECT RANK() OVER(ORDER BY totalamount desc, amountpaid desc) AS Ranking,  
BillID, TotalAmount FROM billing;
```

Ranking	BillID	TotalAmount
1	3	1500.00
2	9	1300.00
3	1	1200.00
4	6	1100.00
5	2	950.00
6	10	900.00
7	4	800.00
8	7	700.00

The Highest amount payable by the patients is Rs. 1500.00

- **Arrange the Medicines as per their Total Cost from Highest to Lowest.**

Syntax:

WITH Cost AS (

SELECT MedicineName, quantityinstock*unitprice AS TotalCost FROM pharmacy)

SELECT RANK() OVER (ORDER BY TotalCost DESC) AS Ranking,
MedicineName, TotalCost FROM cost;

Ranking	MedicineName	TotalCost
1	Insulin Injection	1250.00
2	Vitamin C 1000mg	336.00
3	Paracetamol 500mg	300.00
3	Antacid Tablets	300.00
3	Amoxicillin 250mg	300.00
6	Cough Syrup	280.00
7	Eye Drops	240.00
8	Antibiotic Cream	210.00
9	Pain Relief Spray	180.00

The most expensive medicine bought is Insulin Injection with a Total Cost of Rs. 1250.00

CONCLUSION:

The Hospital Management System (HMS) is a comprehensive solution designed to streamline and automate the day-to-day operations of a hospital. In the modern healthcare environment, where efficiency, accuracy, and speed are essential, this system plays a crucial role in ensuring that both medical staff and patients benefit from a more organized and responsive healthcare process.

This project integrates multiple essential modules such as Patient Management, Doctor Scheduling, Appointments, Admissions, Discharge, Billing, Pharmacy Inventory, and Reporting. Each module is designed to interact with the others, maintaining consistent and up-to-date data across the entire system. For instance, when a patient is admitted, their records are automatically updated in the admissions and billing modules. Similarly, doctors can view patient history, upcoming appointments, and their schedules in one place, significantly reducing administrative burden.



Thank You.

